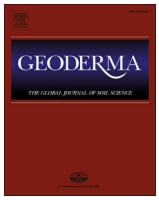
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Systematic review and meta-analysis



Biochar addition reduces salinity in salt-affected soils with no impact on soil pH: A *meta*-analysis

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ABSTRACT

Salinization remains a major issue in soil degradation, for which biochar is a potential solution. In this *meta*-analysis, using 660 paired observations from 99 peer-reviewed articles, we evaluated biochar's effect on salt-affected soils and identified the initial soil properties, biochar properties and experimental factors influencing its efficacy. The results showed that the addition of biochar had a significant ameliorating effect on salt-affected soils. The soil electrical conductivity of saturated paste extract (ECe) was significantly reduced by 13.2%, and such effect was significant in severe salinization soil (rather than slight and moderate salinization). Meanwhile, the soil cation exchange capacity was significantly increased by 17.0%. However, the addition of biochar had no significant impact on soil pH. Model selection analyses further indicated that the level of initial salinity and the type of biochar feedstock were the most important factors regulating the response of soil ECe to biochar addition. In summary, while our study highlights the potential of biochar in ameliorating salt-affected soils, particularly in severely salinized soils, it also underscores the need for more comprehensive research in this field. Additional research is necessary to comprehensively address the significant heterogeneity of biochars, including their thorough characterization. In addition, more studies are required to explore the impact of biochar on both salt-tolerant and non-salt-tolerant plants in salt-affected soils.

1. Introduction

Approximately 1–10 billion ha of soils worldwide are salt-affected (Jesus et al., 2015), including saline, sodic, and saline-sodic soils (Ghassemi et al., 1995). Soil salinization is caused by both natural and anthropogenic factors (Bateman and Muñoz-Rojas, 2019; Oster and Shainberg, 2001; Singh, 2021). Natural processes such as weathering, mineral dissolution, and geological events gradually release salts into the soil profile over time (White, 2005); however, human activities, such as improper irrigation practices, have significantly accelerated the rate of soil salinization (Kitamura et al., 2006; Majumdar, 2001). Additionally, this issue has further been exacerbated by changes in precipitation patterns and rising temperatures induced by climate change (Okur and Örçen, 2020).

Salt-affected soils are characterized by salt stress, high pH, depleted organic carbon (OC), and low nutrient availability (Amanullah et al., 2007; Gong et al., 2021), which reduces productivity (Sahab et al., 2021) and the ability to support crop growth (Ferreyra et al., 1997). The area of saline soils has been expanding (Ivushkin et al., 2019), especially in arid climate zones (Hassani et al., 2020). Eynard et al. (2006) estimated that the global salt-affected soils are increasing by over 1.5 million ha annually, with half of the available arable land at risk of salinization by the mid-century (Sharma and Singh, 2019). Therefore, in a frequently changing global climate, effective measures are warranted to improve salt-affected soils and sustainable agriculture.

However, improving and managing salt-affected soils is challenging (Horneck et al., 2007). Different strategies have been implemented to address these limitations, including hydraulic, chemical, and phytoremediation techniques (Qadir et al., 2000). Irrigation salt washing is a common hydraulic improvement method. Li and Kang (2020) improved the irrigation pattern of local saline lands with drip irrigation technology and reported significant improvements in productivity and vegetation ecology. Zhao et al. (2018) used flue gas desulfurization gypsum as an inorganic material to improve saline soils and observed a decrease in electrical conductivity (EC) after one year. Additionally, halophytes have been reported to improve saline soils (Egamberdieva et al., 2019;

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Hasanuzzaman et al., 2014). Salinity stress can be mitigated by various functional mechanisms in plants (Xia et al., 2019); since most halophytes and salt-tolerant plants have evolved specific mechanisms of osmoregulation or salt secretion (Byrt et al., 2018), they can grow in high salinization soil (Ruan et al., 2010) and absorb soil salt through the root system (De Villiers et al., 1995). However, these methods have shortcomings. For example, the high costs and energy consumption rate of hydraulic engineering hinder its large-scale application for managing salinization (Qadir and Oster, 2004). Moreover, chemical amendments represented by gypsum are likely to have no positive effect (Birru et al., 2019) and are prone to secondary contamination, thereby increasing ecological and environmental risks (Rantamo et al., 2022). Similarly, phytoremediation requires a long time and specific conditions to achieve desired restoration results (Dupae et al., 2014; Fasani et al., 2018).

Biochar, produced by pyrolysis of agricultural waste biomass, has attracted attention as a potential strategy to improve soils (Beesley et al., 2010; Schmidt et al., 2021; Wang and Wang, 2019) and effectively mitigate the environmental risks associated with other techniques (Ippolito et al., 2012). Biochar can improve soil fertility by increasing the accumulation of organic carbon content (Chagas et al., 2022), the biological availability of nitrogen, phosphorus, and potassium (Biederman and Harpole, 2013; Gao et al., 2019), and altering the structure and composition of soil microbial communities (Li et al., 2020; Zhang et al., 2018). These findings have already been reported by global *meta*-analyses that encompass a diverse range of soil types. Nonetheless, there has been relatively limited attention directed towards saline soils in these analyses. For instance, in the study conducted by Liu et al. (2022), out of a total of 1248 data records, only 51 were specifically relevant to saline soils. Moreover, biochar application on salt-affected soils remains controversial (Murtaza et al., 2021). Sánchez et al. (2022) demonstrated the potential of biochar to improve medium salinity soils and enhance productivity. Several other field studies reported the positive effects of biochar in mitigating salt stress (Lee et al., 2022; Xiao et al., 2022; Yao et al., 2022). In contrast, Dahlawi et al. (2018) and Soothar et al. (2021) argued that the high EC in biochar may increase the risk of salt stress. According to a recent *meta*-analysis, the effects of treating soil with added biochar differ between field and pot studies (Singh et al., 2022), indicating that the ameliorative effect of biochar on soil is influenced by environmental factors.

Some basic chemical properties of soil, such as cation exchange capacity (CEC), pH, and electrical conductivity of saturated paste extract (ECe), influence the transformation, availability, and status of nutrients. Therefore, we conducted a comprehensive *meta*-analysis to determine how salt-affected soil chemical properties change following biochar treatment compared with untreated soil and the factors (e.g., initial soil properties, biochar properties, application rates, with/without plants and experiment type) influencing the effects of biochar on salt-affected soils. The definitions, abbreviations, and units of all soil property variables in the text are listed in Table S1.

2. Materials and methods

2.1. Data collection

We searched the Web of Science and Google Scholar databases for all peer-reviewed articles on biochar improvement of salt-affected soils published between January 1980 and May 2022 using the following main keywords: “biochar or charcoal or BC”, AND “saline soil or sodic soil or saline-sodic soil or saline-alkali soil”. The flow chart of the search is illustrated in Fig. S1.

The studies selection criteria were as follows: (i) The paper included at least one biochar treatment and a control group (no biochar treatment) with no less than three replications of the different treatments; (ii) The application of biochar was implemented in salt-affected soils; (iii) the data in each independent study included at least one of the following soil chemical properties: soil salinity indicator (ECe), pH, CEC; (iv) data were presented in specific values with mean and standard deviation (SD). If only the standard error (SE) was reported, SE was converted to SD using the following formula (1):

$$SD = SE \times \sqrt{n} \tag{1}$$

Where n is the number of replication.

For data reported in graphs, extraction was performed using WebPlotDigitizer 4.2 (Burda et al., 2017).

The final 99 papers included in the analysis were identified, and 660 valid data pairs were obtained. Among them, 247 valid data pairs were obtained from 64 papers on the effect of biochar on soil salinity index, 311 valid data pairs were from 82 papers on the effect of biochar on soil pH, and 102 valid data pairs were from 30 papers on the effect of biochar on soil CEC (More details in the Supplementary Material). The locations of the study sites included in this *meta*-analysis are shown in Fig. 1.

2.2. Data treatment

All data were normalized for further comparative analysis. When a paper contained results for multiple sampling dates and soil depths, the data for the last sampling time and the uppermost layer of soil were selected. The application dose of biochar was standardized to a percentage (%). For this, a biochar application rate reported in t ha^{-1} was converted to % using the bulk density (BD) and depth data of biochar-applied soil. If soil BD was not mentioned in the paper, a constant value of 1.3 g cm^{-3} was used. Similarly, if the depth of application was not mentioned in the paper, 20 cm was used.

In addition, soil salinity was generally assessed using ECe (Richards, 1954). However, in the laboratory, EC is usually determined by different soil-to-water mass ratios, such as 1:1, 1:2.5, or 1:5, which were uniformly converted to ECe.

For $EC_{1:5}$, the following equation proposed by Rengasamy (2006) was used to convert it uniformly to ECe:

$$ECe = (14 - 0.13 \times clay\%) \times EC_{1:5} \tag{2}$$

where *clay* represents the clay content (%) of soils. If the texture of the soil was not indicated, we determined the texture of the soil at a similar location based on the World Soil Information Service (WoSIS) soil profile database (Batjes et al., 2017) and selected the average content of clay particles at that texture.

For $EC_{1:1}$ or $EC_{1:2.5}$, depending on the texture of the soil, the following equations proposed by Sonmez et al. (2008) were used to convert them uniformly to ECe. For sandy soil:

$$ECe = EC_{1:1} \times 2.24, \quad ECe = EC_{1:2.5} \times 4.41$$

For loamy soil:

$$ECe = EC_{1:1} \times 2.06, \quad ECe = EC_{1:2.5} \times 3.96$$

For clay soil:

$$ECe = EC_{1:1} \times 1.96, \quad ECe = EC_{1:2.5} \times 3.75 \tag{3}$$

2.3. Data grouping

To allow for cross-comparison between studies, moderators (biochar properties, degree of soil salinity, and experimental conditions) were grouped according to the information published in peer-reviewed articles:

(i) Biochar application rate: low (≤1 %), medium (1–5 %), and high (≥5%).
(ii) Biochar feedstocks were categorized as follows: wood biochar, straw biochar, shell biochar, manure biochar, and mixed biochar (a mixture of ≥2 categories).

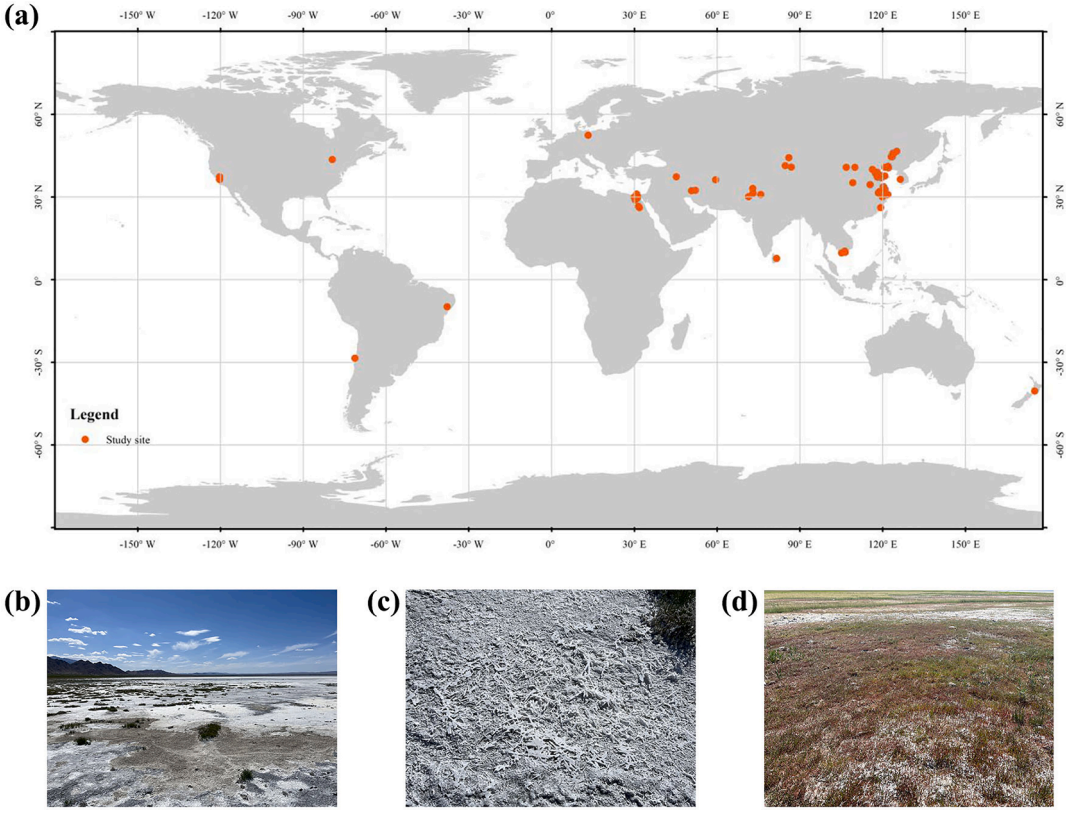


Fig. 1. (a) Global distribution of study sites in this *meta*-analysis, (b) dry lake bed with saline-affected soil and limited vegetation, (c) salt-affected soils with precipitated salt crystals, (d) salt-affected soils supporting the growth of *Salicornia europaea* L. (Photograph credit: Xiao Wang; Location of the photograph: Xinjiang, China).

(iii) Biochar pyrolysis temperature was divided into ≤400 ℃, 400–500 ℃, > 500 °C.
(iv) Experiment types were divided into three categories: field, pot and incubation experiments.
(v) Experimental setups were categorized into two groups based on the inclusion of plants: with plants and without plants.
(vi) Salt-affected soils were classified according to Allison (1954) as follows: slight salinization (ECe: 4–8 dS m^{-1}), moderate salinization (ECe: 8–16 dS m^{-1}), and severe salinization (ECe: > 16 dS m^{-1}).

2.4. Meta-analysis

In this study, the effect size of the applied biochar treatment was quantified by calculating the log response ratio (ln*R*) (Hedges et al., 1999),

$$\ln R = \ln\left(\frac{\overline{X_t}}{\overline{X_c}}\right) = \ln(\overline{X_t}) - \ln(\overline{X_c}) \tag{4}$$

where $\overline{X_t}$ and $\overline{X_c}$ are the mean values of a certain indicator of soils applied with biochar (treatment group t) and without biochar (control group c), respectively.

The corresponding sampling variance (v_i) of ln*R* was calculated by Equation (5):

$$v_i = \frac{S_t^2}{N_t\overline{X}_t^2} + \frac{S_c^2}{N_c\overline{X}_c^2} \tag{5}$$

where N_t, N_c, S_t and S_c are the sample amounts (N_t, N_c) and standard deviations (S_t, S_c) of the treatment group (t) and control group (c), respectively, for a given indicator of soil.

To resolve the problem of non-independence of data caused by multiple effect sizes sharing a common control in a single study, the following methods were utilized. Firstly, we used the variance-covariance matrix (VCV) as the final variance based on the approach of Lajeunesse (2011).

$$V^{\overline{Xc}} = \begin{bmatrix} \frac{(S_c)^2}{N_c\overline{X}_c^2} + \frac{\left(S_t^A\right)^2}{N_t^A\left(\overline{X}_t^A\right)^2} & \frac{(S_c)^2}{N_c\overline{X}_c^2} \\ \frac{(S_c)^2}{N_c\overline{X}_c^2} & \frac{(S_c)^2}{N_c\overline{X}_c^2} + \frac{\left(S_t^B\right)^2}{N_t^B\left(\overline{X}_t^A\right)^2} \end{bmatrix} \tag{6}$$

Secondly, we employed a nested model using the "rma.mv" function from the "metafor" package (Viechtbauer, 2010), where the observation ID was nested within the study as a random effect, to calculate the overall effect size ($\ln R_{++}$) (Nakagawa et al., 2017). If the 95 % CI of the overall effect sizes of an indicator of soil overlaps with "0", biochar is then considered to have no significant effect on the indicator; conversely, if the 95 % confidence interval of the overall effect value did not overlap with "0", then the effect of biochar on the indicator is considered statistically significant (Rosenberg et al., 1997).

To ease interpretation, the overall effect size was eventually converted to a mean percentage change based on the control group:

$$Percentage\ change(\%) = \left(e^{\ln R_{++}} - 1\right) \times 100\% \tag{7}$$

where $\ln R_{++}$ represents the overall effect size.

We employed a mixed-effects *meta*-regression model to examine the relationship between effect sizes and explanatory variables, including initial soil properties (salinity levels and pH), biochar attributes (feedstock, pyrolysis temperatures and pH), and experimental factors (application dose, experiment type and with/without plants). The

significance of explanatory variables in the *meta*-regression was assessed by calculating the heterogeneity statistic (Qm), with a p-value of < 0.05 indicating their contribution to the effect size heterogeneity. Additionally, to assess the relative importance of multiple explanatory variables on salt-affected soil properties following biochar treatment, we utilized the *glmulti package* (Calcagno and de Mazancourt, 2010) for model selection. The importance of each explanatory variable was determined by calculating the sum of Akaike weights obtained from the model selection using Akaike's Information Criteria (AIC). To differentiate between significantly important predictors and non-essential ones in the model, we applied cutoffs of 0.8.

The *meta*-analysis results were assessed for publication bias using the funnel plot method (Philibert et al., 2012). Egger's regression analysis was used to quantify the symmetry of the funnel plots, where $p > 0.05$ indicated good symmetry of the funnel plots (Egger et al., 1997). The observed results suggested the absence of publication bias (Fig. S2). We also conducted a leave-one-out sensitivity analysis to check the reliability and robustness of the overall effect size, and the results showed that omitting any single data point did not affect the results (Fig. S3). All statistical analyses and graphs were conducted with R 4.1.2.

3. Results

3.1. Overall effects of biochar application

Overall, compared to the control, the application of biochar had a significant impact on the soil ECe, leading to a substantial reduction of 13.2 % (Fig. 2a), and significantly increased CEC by 17.0 % (Fig. 2c). In contrast, biochar had no significant effect on pH (Fig. 2b).

3.2. The influence of initial soil properties on the effectiveness of biochar in salt-affected soils

The effect of biochar on soil ECe varied between initial soil salinity levels ($p < 0.05$) (Table 1). We used subgroup analysis to evaluate the effect of biochar in different salinization levels; specifically, biochar significantly reduced soil ECe by 21.5 % under severe salinization level, while the effect of biochar under other salinization levels was not significant (Fig. 3a). Furthermore, the heterogeneity analysis revealed no significant impact of soil salinity levels on soil pH and CEC ($p > 0.05$) (Table 1).

The *meta*-regression analysis indicated that the initial soil pH was not significantly correlated with the effect sizes of soil ECe and CEC ($p > 0.05$) (Fig. 4ac), but negatively significantly correlated with the effect sizes of soil pH ($p < 0.05$) (Fig. 4b).

Table 1
Subgroup heterogeneity test of biochar effect on salt-affected soil amendment.

Factor	Parameter	df	Qm	p-value
Biochar feedstock	ECe	4	14.5556	**0.0057**
	pH	4	1.2788	0.8650
	CEC	2	4.7572	0.0927
Biochar pyrolysis temperatures	ECe	2	2.6507	0.2657
	pH	2	11.7796	**0.0028**
	CEC	2	3.5304	0.1712
Biochar application rate	ECe	2	0.1470	0.9291
	pH	2	31.8021	**<0.0001**
	CEC	2	2.0162	0.3649
Initial soil salinization level	ECe	2	8.4221	**0.0148**
	pH	2	0.3460	0.8411
	CEC	2	0.8854	0.6423
With/without plants	ECe	1	3.3568	0.0669
	pH	1	2.1745	0.1403
	CEC	1	0.6728	0.4121
Experiment type	ECe	2	1.4462	0.4852
	pH	2	4.6086	0.0998
	CEC	2	3.0912	0.2132

Note: df represents degrees of freedom; Q_m represents the heterogeneity caused by an explanatory variable; Q_m corresponding to $P < 0.05$ indicates a significant differential effect of this explanatory variable on an indicator of salt-affected soil.

3.3. The influence of biochar properties and experimental factors on the effectiveness of biochar in salt-affected soils

For the different biochar properties subgroup analyses, biochar feedstock significantly influenced the effect sizes of biochar treatment on soil ECe ($p < 0.05$) (Table 1). Compared with the control, manure biochar caused the largest significant reduction in the soil ECe (24.3 %), followed by straw biochar (21.6 %) (Fig. 3a).

Regardless of the pyrolysis temperature, biochar significantly increased soil CEC (Fig. 3c). However, soil pH was significantly increased by 1.3 % only with the application of biochar produced at 400–500 °C (Fig. 3b). The heterogeneity analysis revealed a significant

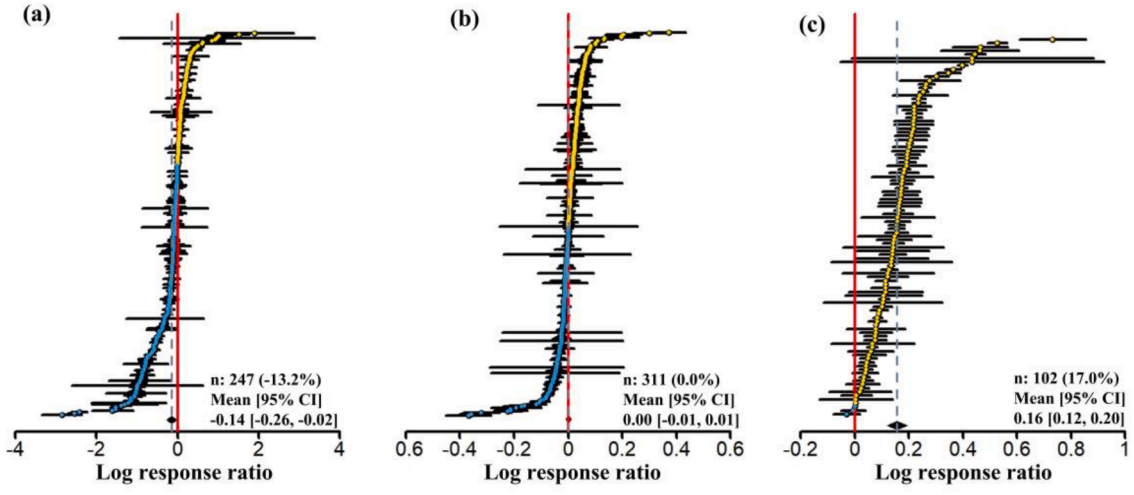


Fig. 2. Overall effect sizes of biochar application on soil (a) ECe, (b) pH, (c) CEC. Mean, overall effect sizes; n, the number of samples, the numbers in parentheses on the right indicate the relative percentage change in the properties of soils affected by biochar; 95 %CI, 95 % confidence intervals.

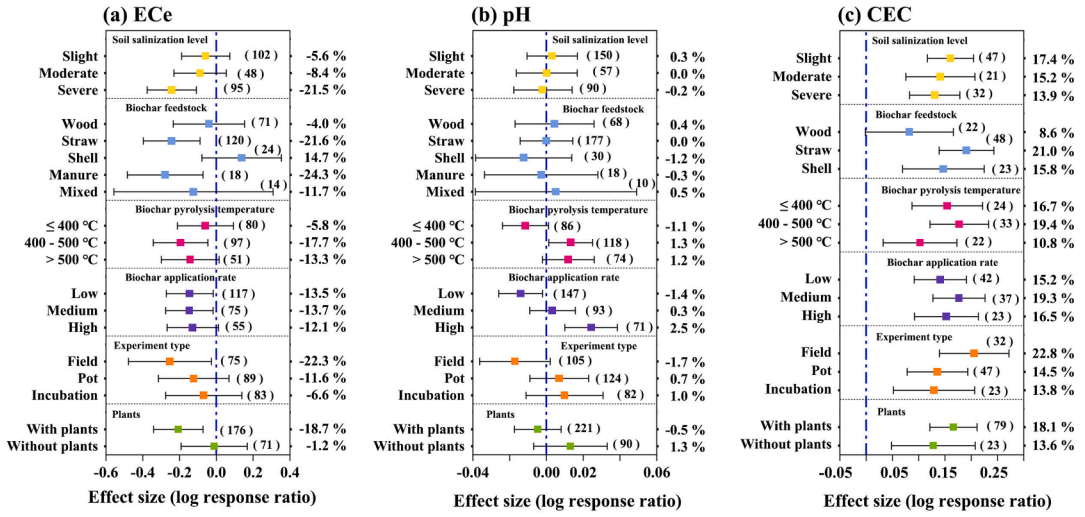


Fig. 3. Overall effect sizes of soil (a) ECe, (b) pH, and (c) CEC under different categories of explanatory variables. Squares and bars represent overall effect sizes and 95% confidence intervals, respectively. The numbers in parentheses indicate the number of samples. The numbers on the right axis indicate the relative percentage change in the basic physicochemical properties for soils affected by biochar.

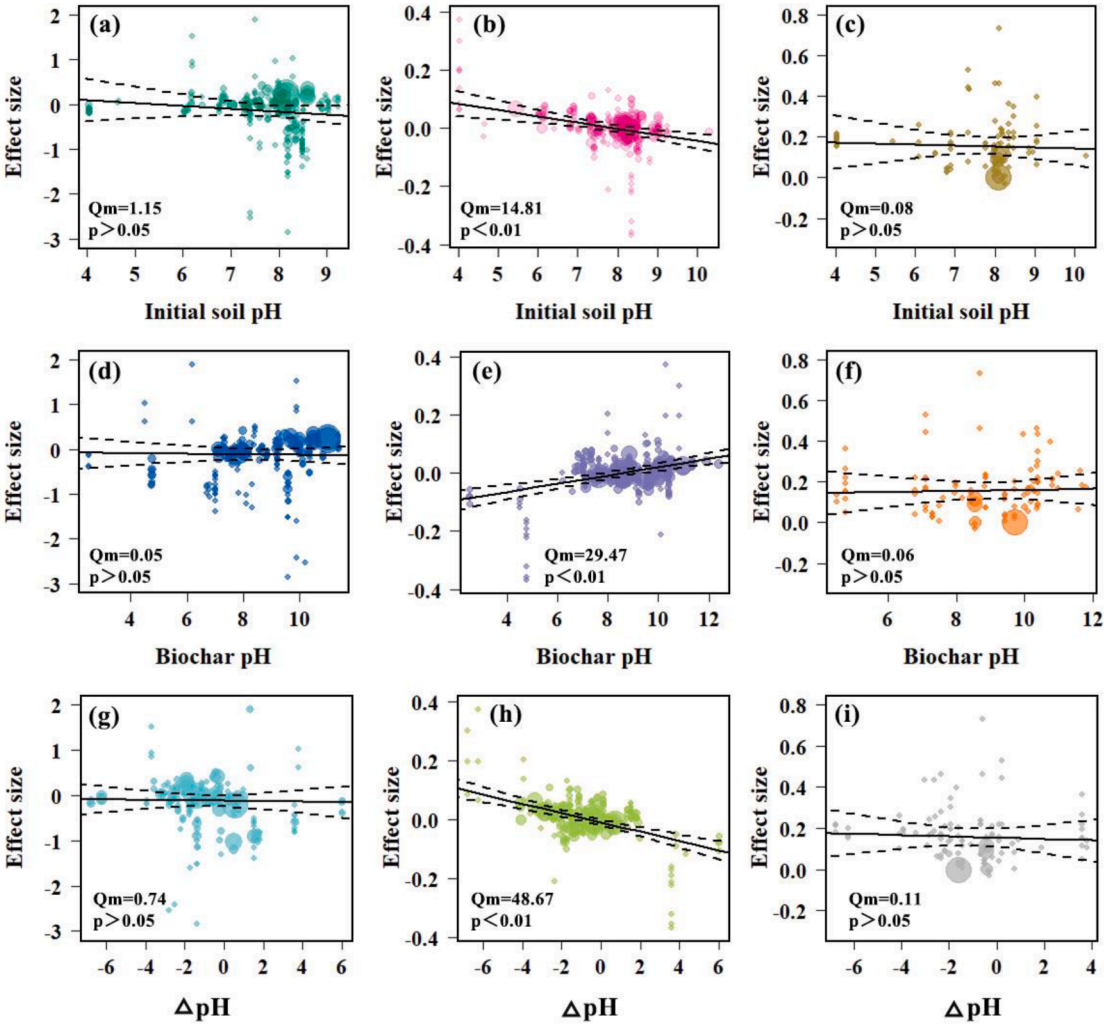


Fig. 4. Effects of initial soil pH on effect sizes (ln*Rs*) of soil (a) ECe, (b) pH and (c) CEC; Effects of biochar pH on ln*Rs* of soil (d) ECe, (e) pH and (f) CEC; Effects of the pH difference between initial soil and biochar (Δ pH) on ln*Rs* of soil (g) ECe, (h) pH and (i) CEC. Black lines are fitted by partial-linear regressions with 95% confidence intervals by dotted lines. Circle sizes correspond to each observation's weight; significance (p values) is shown for each indicator tested.

impact of biochar pyrolysis temperature levels on soil pH ($p < 0.05$) (Table 1).

The *meta*-regression analysis showed that the effect sizes of soil pH was positively correlated with biochar pH and negatively with the pH difference between initial soil and biochar (Δ pH) ($p < 0.05$) (Fig. 4e and h). However, the effect of biochar on soil ECe and CEC did not vary significantly with these factors ($p > 0.05$) (Fig. 4d, f, g and i).

Biochar application rate showed significant differences in soil pH ($p < 0.05$) (Table1), with the increase in biochar application rate, the variation in soil pH changed from a decrease to an increase (Fig. 3b). However, the application of different amounts of biochar did not show significant differences in soil ECe and CEC ($p > 0.05$) (Table 1).

In addition, the heterogeneity analysis showed that there were no significant differences ($p > 0.05$) in the effects of biochar on soil ECe, pH, and CEC among different experiment types or with/without plants (Table 1). It's worth noting that, regardless of the type of experiment, there was no significant change in soil pH (Fig. 3b), but in field experiment type, biochar significantly reduced soil ECe by 22.3 % (Fig. 3a). On the other hand, the observations from the biochar treatments involving plants also showed significant effects on soil ECe (Fig. 3a).

3.4. Relative importance of all explanatory variables

The model selection analysis indicated that the type of biochar feedstock and the initial soil salinization level were crucial predictors for the significant impact of applying biochar on soil ECe (Fig. 5a). For soil pH, the important factors include biochar feedstock, experiment type, biochar application rate and pH, and with/without plants (Fig. 5b). In contrast, soil CEC could not be accurately predicted by any explanatory variables (Fig. 5c).

4. Discussion

4.1. The effect of biochar on salt-affected soil: Decreasing ECe and increasing CEC

The effectiveness of biochar in reducing soil salinity is attributed to various mechanisms. On the one hand, biochar directly reduces soil bulk density through its great porosity and specific surface area (Da Silva Mendes et al., 2021), and promotes soil aggregation structure formation through complexation reaction (Liang et al., 2008) and by increasing humus content (Wang et al., 2014), thereby increasing aeration and permeability (Chaganti and Crohn, 2015) to leaching of soluble salts out of the soil profile (Lashari et al., 2015). Furthermore, in field conditions, these effects are amplified. Environmental factors like rainfall and irrigation expedite the removal of soil soluble salts (Xiao et al., 2020). This might partly explain why the results observed in field studies are better than those in laboratory studies (Fig. 3a). Laboratory studies are often conducted in controlled environments, while field studies take into account more natural and practical agricultural factors (Bresler et al., 2012; Greggio et al., 2012). On the other hand, the strong adsorption capacity of biochar (Siedt et al., 2021) enables it to effectively adsorb and immobilize soil salts on its surface or within its pores (Duan et al., 2021; Singh et al., 2021), thereby helping to reduce the salt concentration in soil solutions. Additionally, some biochars have a considerable concentration of calcium (Joseph et al., 2021; Singh et al., 2010), which could solubilize in soil and facilitate the removal of excess exchangeable sodium from cation exchange sites (Sadegh-Zadeh et al., 2018; Ullah et al., 2022; Xiao and Meng, 2020). Lastly, biochar has been shown to affect plant roots significantly (Jabborova et al., 2021). Biochar enhances the root vigor of plants in salt-affected soils, improving hydraulic conductivity and creating water channels more favorable to salt-leaching loss (Saifullah et al., 2018).

Previous studies have shown similar results for other organic additives. For instance, a *meta*-analysis by Li et al. (2023) found that the application of organic material reduced soil ECe by approximately 11.8 %. Similarly, Chen et al. (2020) reported that grass as an organic material decreased EC by 10.1 %. Oo et al. (2015) also found a decrease in EC due to compost application, and Lakhdar et al. (2009) noted that this reduction in ECe was mainly attributed to the increased leaching of salts by compost. In our *meta*-analysis indicated that biochar reduced ECe by 13.2 % (Fig. 2a). While biochar demonstrated a potentially stronger impact in reducing soil ECe compared to conventional organic additives, its effectiveness still aligns with the broader range observed for these additives. We hypothesize that the significant effects of biochar may stem from its unique physicochemical properties, microbial conditioning, sorption capability, and soil structure improvement mechanisms (Amini et al., 2016; Mona et al., 2019; Siedt et al., 2021), which other organic additives lack. Due to our limited understanding of other organic amendments, further research is warranted to explore their role in decreasing ECe in salt-affected soils and to compare them with biochar.

It is worth noting that while biochar generally led to a reduction in soil ECe, we did observe certain cases with contrasting trends (Fig. 3). This phenomenon could be attributed to the characteristics of biochar, particularly in situations where the biochar itself carries a higher salinity (Mavi et al., 2020). Therefore, future studies should explore the characteristics of biochar itself as well as its interrelationship with soil salinity.

Further, biochar treatment increased the amount of cation exchange in salt-affected soils (Fig. 2c). Laird et al. (2010) found that soils with high CEC could retain or bind nutrient elements to biochar particles, humus, and clay surfaces. Meanwhile, biochar has been reported to affect soil properties by altering the quality and quantity of litterfall and root secretions entering the soil (Zhu et al., 2017). The net primary productivity in soil treated with biochar is relatively higher compared to untreated soil, effectively increasing organic carbon content (Mavi et al., 2023).

4.2. The negligible impact of biochar on soil pH

Notably, biochar treatment did not significantly affect soil pH (Fig. 2b). However, a previous *meta*-analysis by Biederman and Harpole (2013) who reported an increase in pH in soils corresponding to biochar

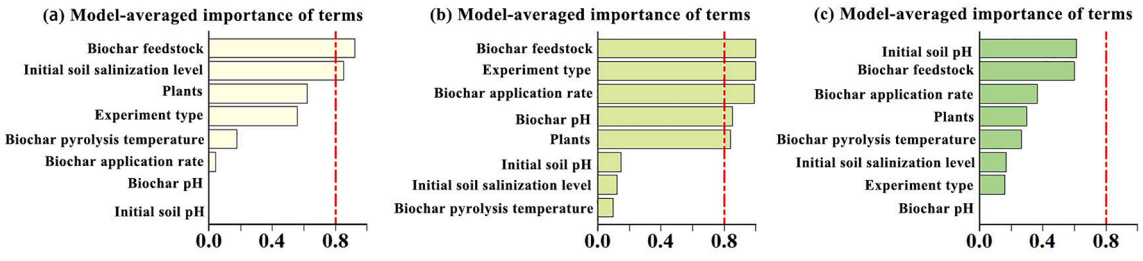


Fig. 5. Average importance of the predictors estimated as a result of model selection analysis, (a) ECe, (b) pH, (c) CEC. The sum of the Akaike weights value was >0.8 (red dotted line), indicating that the predictor variable was of high importance.

application. In our data, most soils were alkaline. Meanwhile, our results indicated that the soil pH improvement depended on the pH gap between soil pH before the experiment and biochar pH (Fig. 4h). As Gao et al. (2021) also mentioned, the match between biochar pH and soil pH has a significant impact on the efficiency of biochar in enhancing the plant photosynthetic rate. Hailegnaw et al. (2019) showed that the effect of biochar on pH was more significant when the soil pH was close to the acidic range. Salt-affected soils typically have a high initial pH (Chhabra, 2004; Gupta and Gupta, 2014), which may limit the effect of biochar on pH. Another possible explanation is that salt-affected soils usually have a high buffering capacity to resist pH changes caused by external factors such as biochar (Zheng et al., 2018).

4.3. Influential factors in soil salinity reduction

4.3.1. Biochar feedstock

The properties of biochar modulate its effect on the salinity and other chemical properties of salt-affected soils. Notably, biochar feedstock is a key factor affecting soil ECe following biochar treatment (Fig. 5a). This suggests that using a particular biochar feedstock may lead to a greater reduction in ECe levels. As reported by Mavi et al. (2020), the addition of biochar from different feedstocks to salt-affected soils reduced ECe to varying degrees. The variation in effectiveness may be attributed to differences in factors such as biochar carbon, nitrogen, and phosphorus concentrations, pH, functional groups (e.g., –COOH and $–NH_2$), nutrients (Mg^{2+} or Ca^{2+}), surface morphology, and structure (Ali et al., 2017; Haider et al., 2022; Li and Li, 2022; Qian et al., 2023).

Surprisingly, typical manure biochar has a high salt content (Zhao et al., 2013); furthermore, this *meta*-analysis found that manure biochar reduced ECe more than other feedstock biochars (Fig. 3a). Ippolito et al. (2020) and Su et al. (2021) showed that manure biochar contains high mineral nutrients, such as calcium (Ca^{2+}) and magnesium (Mg^{2+}), which are involved in ion regulation and neutralization in the soil, thus reducing the salt concentration. Moreover, according to the study conducted by Mavi et al. (2020), manure biochar demonstrates a greater ability compared to other feedstock biochars in enhancing the survival capacity of soil microbial biomass under saline conditions. However, it is important to note that there may be certain limitations in the subgroup analysis of biochars from different feedstocks in this study. The limited number of comparisons in some feedstock categories could potentially affect the reliability of the subgroup analysis. Therefore, although we propose some explanations, its exact mechanism must still be explored more thoroughly in future studies.

4.3.2. The level of initial salinization

The difference in the effectiveness of biochar in amending salt-affected soils was attributed to the degree of salinization of the initial soil (Mahmoud et al., 2020; Yao et al., 2021). The results of the model selection indicated that the degree of soil salinity was an important factor influencing ECe in the amended soil (Fig. 5a). Biochar showed the best reduction of soil ECe in severe salinization (Fig. 3a). In heavily salinized soils, these higher salts can be adsorbed/retained on the surface or trapped in the fine pores of the biochar to reduce the accumulation of salts in the soil surface layer (Liang et al., 2021; Saifullah et al., 2018). However, soil salinity is too high above a certain threshold and the effectiveness of biochar may be limited. In future research, it is essential to delve deeper into the critical range of soil salinity to determine the optimal application conditions for biochar.

5. Conclusions and insights for future research

We present the global-scale *meta*-analysis to summarize the effect of biochar application on the improvement of salt-affected soil. Our findings indicated that biochar application significantly altered ECe and CEC of salt-affected soils, with the exception of soil pH. Among these changes, biochar feedstock and salinization level were the key variables that determined the response of salt-affected soil ECe to biochar. This study provided evidence that biochar is a potential soil amendment for salt-affected soils. However, our study has some limitations. For example, when conducting subgroup analyses, we removed subgroups with small sample sizes, which limited the data on the effects of factors, such as biochar properties, on restoration effectiveness. Future research should concentrate on a detailed characterization of biochars, including aspects such as the hydrogen to carbon (H:C) ratios, electrical conductivity, and types of pyrolysis. In addition, our analysis focused on the differences in the effectiveness of biochar in improving salt-affected soils under conditions with and without plants, can build upon this foundation to further explore the efficacy of biochar under different types of plants in future, such as evaluating the impact of biochar on salt-tolerant and non-salt-tolerant plants. In summary, in future remediation efforts, it is essential to fully consider biochar properties, initial soil properties, and experimental factors to optimize improvement strategies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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